# MILITARY SPECIFICATION

## SLIP RING ASSEMBLY

This Specification is approved for use within the US ArmY Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Purpose. This specification establishes performance, test, manufacture, design and acceptance for the sliP ring assembly for use in the Vulcan Air Defense System (VADS).

1.2 Item description. The slip ring assembly transfers electrical energy from a stationary structure to a rotating structure.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and anypertinentdata whichmaybeof useinimprovingthis documentshouldbe addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using theself-addressed Standardization Document Improvement Proposal (DD Form 1426) appearingat the end of this document orby letter.

AMSC N/A FSC 1005
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:51Z
Check the source to verify that this is the currentversion before use.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

P8437092 - Packaging Data Sheet
8437092 - Slip Ring Assembly

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 order of precedence. In the event of a conflict between the text of this specification and the referenced specifications cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Physical characteristics.

3.1.1 Weight. The weight of the slip ring assembly shall be no greater than 16 pounds.

3.2 Performance requirements.

3.2.1 Circuit description. 28 circuits shall be provided as follows:

a. 24 circuits - 1 ampere, 28V DC.

b. 2 circuits - 100 amperes resistive load, 28V DC.

c. 2 circuits - RF.



Source: https://assist.dla.mil -- Downloaded: 202610-23T03:51Z
Check the source to verify that this is the current version before use.

3.2.2 Rotational speed. The slip ring assembly shall be capable of operation at speeds up to 20 RPM.

3.2.3 Breakaway torque. The torque required to initiate movement of the rotor relative to the stator shall not exceed 30 inch-pounds.

3.2.4 Noise. Slip ring noise shall not exceed 400 microvolt at a test current of 50 milliamps during rotation at 20 RPM for each ring pair.

3.2.5 Dielectric strength. The slip ring assembly shall withstand a voltage of 500V DC for 10 seconds between each circuit and all other circuits and case ground with no evidence of arcing or breakdown.

3.2.6 Insulation resistance. Insulation resistance between each circuit and all other circuits and case ground shall be not less than 50 megohms at 250V DC.

3.2.7 Circuit resistance. The circuits shall not exceed the following resistances: 1 ampere signal circuits, 110 milliohms; 100 ampere power circuits, 5 milliohms; and RF circuits, 50 milliohms.

3.2.8 Voltage standing wave ratio. The voltage standing wave ratio of the RF rings shall be 1.40 maximum at frequencies of 30 to 76 megahertz (MHz).

3.2.9 Crosstalk. Crosstalk drop between channels of RF slip rings shall be 50db minimum at 30 to 76 MHz.

3.2.10 Waterproofness. The slip ring assembly shall show no signs of leakage when pressurized internally to 3 PSI and submerged in water to a depth of 12 inches from the top of the unit for a period of 5 minutes.

3.3 Design and construction. The slip ring assembly shall be fabricated and assembled in accordance with dwg. 8437092 and associated drawings.

3.3.1 Materials. Materials used in construction of the slip ring assembly shall be in accordance with dwg. 8437092 and associated drawings.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

3.3.2 Protective coating. Protective coating used in construction of the slip ring assembly shall be in accordance with dwg. 8437092 and associated drawings.

3.3.3 Product marking. The slip ring assembly shall be marked in accordance with Dwg. 8437092. C o nnector designations shall be marked in accordance with dwg. F8437092.

3.3.4 Workmanship. Workmanship shall be in accordance with MIL–STD-454.

3.5 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified, in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

| Part description | Drawing | Quantity |
|---|---|---|
| Slip ring assembly | 8437092 | 2 |

4.3.2 Inspection to be performed. First article slip ring assemblies shall be subjected to any or all of the examinations and tests specified in Table I and to any or all requirements of the applicable drawing.

4.3.3 Rejection. The first article quantity shall be rejected upon failure of any slip ring assembly in the sample to comply with any visual or dimensional characteristics or upon failure to comply with the criteria for any test. The Government reserves the right to terminate its inspection whenever a slip ring assembly fails to comply with any visual or dimensional characteristics or slip ring assembly fail to comply with the criteria for any test.

4.4 Quality conformance inspection. Each slip ring assembly submitted for acceptance under contract shall be subjected to the acceptance inspections specified herein. Failure of any slip ring assembly to conform to any acceptance inspection requirement shall be cause for rejection of that unit.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MI

| PARAGRAPH | TITLE | SHEET | DRAW |
|---|---|---|---|
| | Slip Ring Assembly | 1 OF 1 | See |
| | | | NEXT |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARA |
|---|---|---|---|---|---|
| | Slip Ring Assembly (dwg. 8437092) | | | | |
| | Breakaway Torque | 2 | | 3.2.3 | 4. |
| | Noise | 2 | | 3.2.4 | 4. |
| | Dielectric Strength | 2 | | 3.2.5 | 4. |
| | Insulation Resistance | 2 | | 3.2.6 | 4. |
| | Circuit Resistance | 2 | | 3.2.7 | 4. |
| | Voltage Standing Wave Ratio | 2 | | 3.2.8 | 4. |
| | Crosstalk | 2 | | 3.2.9 | 4. |
| | Waterproofness | 2 | | 3.2.10 | 4. |
| | Weight, max | 2 | | 3.1.1 | St an |
| | Poor Workmanship | 2 | | 3.3.4 | 4. |

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

TABLE II. QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MI

| PARAGRAPH | TITLE | | | SHEET 1 OF 1 | DRAWIN 843 |
|---|---|---|---|---|---|
| 4.4.2 | Slip Ring Assembly | | | | NEXT |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGR |
| Critical | None defined | | | | |
| Major | | | | | |
| 101. | Breakaway Torque | | 100% | 3.2.3 | 4. |
| 102. | Noise | | 100% | 3.2.4 | 4. |
| 103. | Dielectric Strength | | 100% | 3.2.5 | 4. |
| 104. | Insulation Resistance | | 100% | 3.2.6 | 4. |
| 105. | Circuit Resistance | | 100% | 3.2.7 | 4. |
| 106. | Voltage Standing Wave Ratio | | 100% | 3.2.8 | 4. |
| 107. | Crosstalk | | 100% | 3.2.9 | 4. |
| 108. | Waterproofness | | 100% | 3.2.10 | 4. |
| 109. | Poor Workmanship | | 100% | 3.3.4 | 4. |
| Minor | None defined | | | | |

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, w

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

4.4.1 Inspection eqipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in Tables I and II. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract.

4.5 Methods of inspection.Unless otherwise specified for individual tests, the slip ring assembly shall be subjected to test under the following conditions:

| | |
|---|---|
| Temperature: | 150C to 35°C (59°F to 95°F) |
| Relative Humidity | 45% to 90% |
| Atmospheric Pressure | 650 MMHG to 800 MMHG |

The tests may be performed in any sequence.

4.5.1 Examination of product. The slip ring assembly shall be inspected to verify compliance with MIL-STD-454, requirement 9 and dwg. 8437092.

4.5.2 Breadaway torque. With the slip ring assembly mounted in an upright position and supported by the mounting base, the tangential force required to initiate movement, applied at a known distance from distance from center, shall be measured. The product of the distance to center of rotation in inches and the applied force shall not exceed 30 in-lbs.

4.5.3 Noise. The slip ring assembly shall be mounted in a suitable rotating fixture and the brush leads shorted together. With the shaft rotating at 20RPM and a test current of 50 milliamperes the noise voltage measured across two brush leads shall not exceed 400 microvolt peak to peak.

4.5.4 Dielectric strength. A test potential of 500V DC shall be applied between each circuit and all the circuits and case ground. There shall be no evidence of arcing or breakdown.

4.5.5 Insulation resistance. Insulation resistance shall be tested between each circuit and all others and case ground. At a test voltage of 250V DC, the insulation resistance shall be not less than 50 megohms.

4.5.6 Circuit resistance. The circuits shall be tested for resistance between corresponding terminals on the rotor and stator connectors. The measured resistance shall not exceed the following:

| | |
|---|---|
| Signal circuits: | 110 milliohms |
| Power circuits: | 5 miilionms |
| RF circuits | 50 milliohms |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

4.5.7 Voltage standing wave ratio. The voltage wave ratio of the RF circuits shall be measured at 30 to 76 megahertz. The maximum voltage standing wave ratio shall be 1.4:1.

4.5.8 Crosstalk. Crosstalk drop between channels of the RF circuits shall be measured at 30 to 76 megahertz. The attenuation shall be 50db minimum.

4.5.9 Waterproofness. The slip ring assembly shall be pressurized internally to a pressure of 3 PSI and submerged in water to a depth of 12 inches from the top of the unit for a period of 15 minutes. During this period, the slip ring assembly shall show no signs of leakage. The unit shall then be pressurized internally to 3 PSI below atmosphere. During this period, the slip ring assembly shall show no signs of leakage.

## 5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with Packaging Data Sheet P8437092 unless otherwise specified in the contract.

## 6. NOTES

6.1 Intended use. The slip ring assemblies manufactured under this specification are intended for use in Vulcan Air Defense System (VADS).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of part drawing.
c. Provisions for first article inspection, (see 4.3).
d. Any deviation from this specification.
e. Level of packaging (see 5.1).

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander: US Army Armament, Munitions and Chemical Command, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001. Request letter of submittal shall state contractor contract number, specification number, item nomenclature and classification of defects or test.

6.4 Subject term (keywords) listing.

Slip Ring Assembly
Vulcan Air Defense System

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1005-A664)



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

**INSTRUCTIONS:** In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

*(Fold along this line)*

*(Fold along this line)*

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armament Munitions &
Chemical Command
ATTN: AMSMC-QA
Dover, NJ 07801-5001

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-S-70685 (AR) | Slip Ring Assembly |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:51Z
Check the source to verify that this is the current version before use.